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Sperm whale depredation on longline surveys and implications for the assessment of Alaska sablefish

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ABSTRACT

Sperm whales (*Physeter macrocephalus*) in the Gulf of Alaska depredate (remove or damage fish caught on fishing gear) on the annual National Marine Fisheries Service's longline survey. Depredation can reduce sablefish (*Anoplopoma fimbria*) catch rates and increase uncertainty in survey-derived estimates of sablefish biomass. Using 27 years of longline survey data, this study: 1) evaluates fixed- and mixed-effects generalized linear models to estimate the effects of sperm whale depredation on the sablefish survey abundance; and 2) evaluates the impact of accounting for sperm whale depredation in the sablefish stock assessment. Model evaluation and simulations showed that mixed-effect models were far superior to fixed-effect models in terms of precision and confidence interval coverage. The estimated reduction in sablefish catch rate due to depredation was approximately 15%, which was considerably higher than previous estimates. Correcting for sperm whale depredation in the assessment resulted in a 2% increase in estimated female spawning biomass in the terminal year and a 3% higher quota recommendation, valued at approximately US $3 million. Accounting for sperm whale depredation in the sablefish assessment should be done in concert with estimating the increase in fishing mortality caused by depredation in the commercial fishery.

1. Introduction

Sperm whale (*Physeter macrocephalus*) depredation (whales removing or damaging fish caught on fishing gear) is a key management concern for the National Marine Fisheries Service's (NMFS) longline survey and the Alaska commercial sablefish (*Anoplopoma fimbria*) fishery (Hanselman et al., 2016; Peterson and Carothers, 2013). There have been many studies with observations of mammal depredation on longline fisheries (e.g., Ashford et al., 1996; Secchi and Vaske, 1998; Nolan et al., 2000; Rabearisoa et al., 2015, Dalla Rosa and Secchi, 2007). The majority of the published research has examined interactions with killer whales, and much of the literature is focused on mitigation and deterrence strategies (Werner et al., 2015). However, very few studies outside of Alaska have attempted to quantitatively estimate the effect of depredation on catch-per-unit-effort on longline fisheries or surveys. The several quantitative studies that exist are based on fishery data and use *t*-tests (Purves et al., 2004), GLMs (Clark and Agnew, 2010; Passadore et al., 2015), or GLMMs (Tixier et al., 2016).

Sperm whale depredation, specifically, is a global issue that has also been documented in the Southern Ocean, associated with Patagonian toothfish (*Dissostichus eleginoides*) fisheries (Ashford et al., 1996; Purves et al., 2004; Roche et al., 2007; Tixier et al., 2010), and in North Atlantic Ocean fisheries for Greenland halibut (*Reinhardtius hippoglossoides*; Dyb, 2006; Mesnick et al., 2006). This study evaluates several statistical models for estimating sperm whale depredation on the NMFS longline survey and examines the implications of including sperm whale depredation in the sablefish stock assessment.

Sperm whale depredation on the commercial fishery and NMFS longline survey generally occurs in the central and eastern Gulf of Alaska (GOA), impacting sablefish and Pacific halibut (*Hippoglossus stenolepis*) catches during haulback of demersal longline fisheries. Observations of depredation are relatively recent, starting in the mid-1980s for the commercial longline fishery in Alaska and enumerated since 1998 for the NMFS longline survey (Straley et al., 2014). In contrast to killer whale (*Orcinus orca*) depredation in western Alaska, which has a strong negative effect (54–72%) on catch per unit effort (CPUE) (Peterson et al., 2013, 2014), the most recent study of sperm whale depredation on the NMFS longline survey estimated small ($\sim -2\%$) effects on sablefish CPUE (Sigler et al., 2008). Our analysis improves upon that of Sigler et al. (2008) by incorporating observations

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from additional years and utilizing a mixed-effects statistical model to better estimate depredation effects.

Sablefish are a deep-dwelling, commercially valuable species in the northeastern Pacific that have been targeted by domestic and foreign fisheries since the early 1900s (Hanselman et al., 2016; McDevitt, 1986; Sasaki 1985). The 2014 catch of 11,580 t had an ex-vessel value of U.S. $98 million (Fissel, 2014), making sablefish one of the most valuable ($/kg) species in the region. The commercial sablefish fishery in Alaska federal waters has been managed under an Individual Fishing Quota (IFQ) system since 1995. The fishery is open each year from roughly mid-March to mid-November, and NMFS conducts a sablefish stock assessment annually between September and November. The current stock assessment model fits three active abundance indices (the longline survey evaluated in this study, a bottom trawl survey, and the US commercial longline fishery), and two historical abundance indices (the Japanese longline fishery and Japan-US cooperative longline survey; Hanselman et al., 2016). A recommended catch limit for the following year, called the Allowable Biological Catch (ABC), is calculated by applying a target fishing mortality rate to the estimate of present abundance projected for the next year. The ABC is then used to determine IFQ limits for the commercial fishery.

Currently, stations impacted by sperm whale depredation are included in the stock assessment, whereas skates (100 m section of longline with 45 hooks) impacted by killer whales are excluded from abundance calculations because the killer whale effect is easier to detect and typically severe (Peterson et al., 2013). However, there is concern that failing to account for removals of survey sablefish due to sperm whale depredation could result in biased assessment results (Hanselman et al., 2016). Biases in assessment results would only occur if there was a temporal trend in depredation on the survey, because the survey is used as an index of abundance in the assessment model and catchability is estimated. We show in this study that incidences of sperm whale depredation have increased over time, and hence, it is important to evaluate potential assessment biases due to depredation. However, estimating catch losses attributable to sperm whale depredation can be difficult because depredation does not always leave evidence, such as damaged fish or hooks on the fishing gear (Clark and Agnew, 2010; Peterson, 2014; Tixier et al., 2010). Additional challenges include variable catch rates, the sporadic nature of sperm whale depredation, which creates a highly unbalanced design, and correlation within and among longline survey station data. Several prior studies of whale depredation from this region utilized fixed-effects models (Hanselman et al., 2010; Hill et al., 1999; Straley et al., 2005; Peterson et al., 2013). However, analysis of unbalanced designs using fixed-effects models can result in poor estimation and inference compared to mixed-effects models (Garson, 2012).

Thus, in this study, we first compared the performance of Generalized Linear Mixed Models (GLMMs) with fixed-effects Generalized Linear Models (GLMs) traditionally used for estimating changes in CPUE due to sperm whale depredation, while accounting for covariates such as depth and location. Second, we evaluated the management implications of incorporating the sperm whale depredation effect into the annual Alaska sablefish stock assessment.

2. Materials and methods

2.1. Longline survey data collection

NMFS sablefish longline survey stations in the GOA have been sampled every year from June to August, 1990–2016. Survey stations generally align with sablefish commercial longline fishing grounds along the continental slope and are systematically spaced approximately 30–50 km apart (Fig. 1) (Peterson et al., 2013; Sigler et al., 2008). In a given year, each station was fished for one day from shallow to deep (depths ranging from roughly 150–1000 m) using two sets hauled end to end (Peterson et al., 2013). Each set consisted of 80 skates (string of 45 hooks), providing a total of 160 skates (7200 hooks) fished per station. Depth was recorded every fifth skate and interpolated for all other skates. Hooks were spaced 2 m apart and baited with squid. Upon retrieval, hooks were deemed "ineffective" if they were straightened, snarled, bent, or in any way unable to fish correctly (Hanselman et al., 2016; Peterson et al., 2013). While it is conceivable that some "ineffective" hooks could be caused by whale depredation, the proportion of ineffective hooks in the data was small (1.8%) and because these hooks are excluded in the abundance calculations, we excluded them in our analysis as well. Gully stations, which are stations that sample shallower cross-shelf habitat, were excluded because most are not used in the abundance index, are rarely affected by whale depredation, and are only weakly correlated with overall trends in CPUE (Hanselman et al., 2016). At each station, sablefish catch, effort, age, length, weight, and maturity data were collected (Hanselman et al., 2016).

Indicators of potential sperm whale interactions with the longline survey were collected starting in 1998. Two indicators were tracked at the station level: 1) "presence" of sperm whales (e.g., sightings within 100 m of the vessel); and 2) "evidence" of depredation, when sperm whales were present and retrieved sablefish were damaged in characteristic ways (e.g., missing body parts, crushed tissue, blunt tooth marks, shredded bodies). Damaged fish caused by other animals (e.g., sea lions and sharks) are rare for both the survey and the fishery. Incidences of sperm whale depredation on the longline survey have been frequent in the Central Gulf of Alaska (CGOA), West Yakutat (WY), and East Yakutat/Southeast (EY/SE) management areas (Fig. 1), but rare in the Bering Sea, Aleutian Islands, and Western Gulf of Alaska (Hanselman et al., 2016; Sigler et al., 2008). These previous studies have examined both the "presence" and "evidence" indicators, but our modeling showed that the "evidence" flag provided stronger inferences, so we largely focus on that indicator in this study.

The longline survey data are used to derive annual estimates of relative population numbers (RPN, an abundance index) for use in the sablefish stock assessment model (Hanselman et al., 2016). The RPN indices are computed by management area across five depth strata considered to be in exploitable habitat (200–300 m, 300–400 m, 400–600 m, 600–800 m, and 800–1000 m). Specifically, sablefish CPUE data are computed for each station and depth stratum by dividing total catch by the number of effective hooks fished. To estimate RPNs by management area, the CPUE data are then averaged across stations, multiplied by strata-specific habitat area sizes, and summed across depth strata.

2.2. Spatial and temporal patterns

Data analysis and modeling were limited to stations in the three areas with prevalent evidence of sperm whale depredation: CGOA (16 stations); WY (8 stations); and EY/SE (11 stations). These three areas contained 98% of the sperm whale depredation events across survey stations (Hanselman et al., 2016). For each area, logistic regression was used to assess time trends in the annual proportion of stations with sperm whale presence or evidence of depredation. To facilitate model development and interpretation, we first assessed patterns of variation in sablefish CPUE. We computed annual means of CPUE by station and standardized each index to have a mean of zero and standard deviation of one across years. The 35 standardized CPUE indices were then used in a principal components analysis (PCA; Dunteman, 1989) to assess the dominant spatial and temporal patterns in CPUE across stations and years. Full results and discussion from the PCA analysis are shown in Supplementary material.

2.3. Modeling depredation

The effect of whale depredation on survey sablefish CPUE was estimated using several model forms. Building upon previous studies

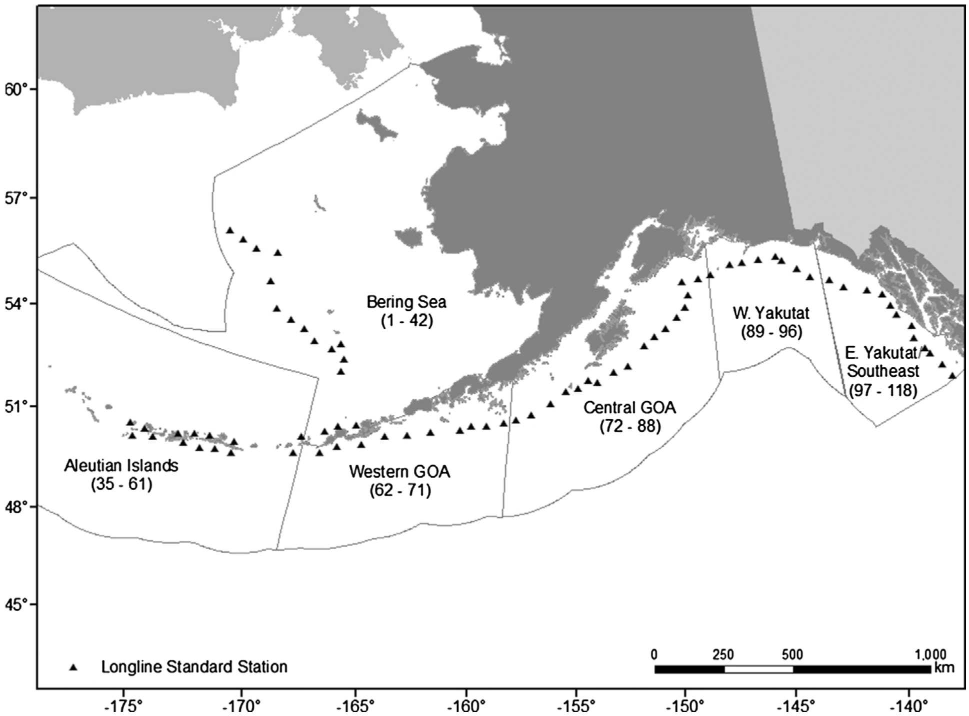

Fig. 1. Slope survey stations in the Gulf of Alaska for the Alaska Fisheries Science Center longline survey. Station identification numbers are noted within each area.

(Clark and Agnew, 2010; Hanselman et al., 2010; Peterson et al., 2013; Sigler et al., 2008; Tixier et al., 2010, 2014), we compared depredation estimates across several GLMs and GLMMs. GLMs are a family of statistical models that can account for non-normal response distributions, such as those for count data. A number of distributions were evaluated to account for overdispersion (when the variance is greater than the mean) in the sablefish catch data (Hilbe, 2011). Distributions considered included Poisson, quasi-Poisson (QP), and negative binomial (NB; accounts for overdispersion by adding an additional parameter to model the higher variance) (Hilbe, 2011; Venables and Dichmont, 2004; Venables and Ripley, 2002; Zeileis et al., 2008; Zuur et al., 2009).

GLMMs are an extension of GLMs that allow a mixture of fixed and random effects (Bolker et al., 2009; Venables and Dichmont, 2004; Zuur et al., 2009). Fixed-effects models evaluate the specific effects of variables at each level, whereas mixed models assume random effects for some variables (often batched or grouped data) where effects are assumed to follow a distribution representing a "population" of potential effects (Bolker et al., 2009; Kinney and Dunson, 2007), accounting for individual-level variation in effect sizes. GLMMs that include random effects can be effective at dealing with unbalanced or non-Gaussian data (Bolker et al., 2009; Venables and Dichmont, 2004; Venables and Ripley, 2002).

A log-linear model of CPUE that accounts for the full structure of the survey data across areas (A_i), years (Y_t), depth strata (D_j), and stations (S_k) is given by:

$$\begin{aligned}\log(C_{tjk[i]}) = \log(H_{tjk[i]}) + A_i + Y_t + D_j + S_{k[i]} + (YA)_{ti} + (YD)_{tj}\\ + (YS)_{tk[i]} + (AD)_{ij} + (DS)_{jk[i]} + (YAD)_{tij} + (YDS)_{tjk[i]}\end{aligned} \tag{1}$$

where the subscript $k[i]$ indicates that station k is nested in area i, C denotes aggregated catch (summed across skates), and H denotes total effective hooks (summed across skates). The term H is a constant that is specified as an "offset" in model fitting (Venables and Ripley, 2002). Eq. (1) is a "fully saturated" model because it includes all main-level effects (Y_t, A_i, etc.) and two-way and three-way interactions up to the level of the aggregate data with $(YDS)_{tjk[i]}$. With Eq. (1) in mind, the following sections outline alternative models used to estimate depredation effects of sperm whales in the GOA.

2.4. Model fitting

Four models were compared for estimating sperm whale depredation effects. We used evidence of depredation (e.g., damaged fish) as the indicator variable used to flag stations for sperm whale depredation, consistent with Sigler et al. (2008). The first two models were fixed-effects models (GLMs) with different distributions (QP and NB). The QP and NB models had the following form, similar to that used by Peterson et al. (2013):

$$\log(C_{tjk[i]}) = \log(H_{tjk[i]}) + A_i + Y_t + D_j + S_{k[i]} + (YS)_{tk[i]} + (AD)_{ij} + \lambda F_{tk}, \tag{2}$$

where the coefficient λ denotes the effect of depredation, and F is an indicator variable for depredation ($F = 1$ when a station is flagged for depredation and $F = 0$ no depredation). The third model (denoted ME.1), also based on the structure in equation (2), was a mixed-effects model assuming a Poisson distribution for C_{tjk}. The terms for year (Y_t), station (S_k), and their crossed interactions $(YS)_{tk}$ were treated as random effects. Each random effects term was assumed to follow a normal distribution, $Y_t \sim \mathrm{N}(0, \sigma_Y^2)$ (Schall, 1991).

The final model (ME.2) was also a Poisson mixed model and had a complete factorial structure:

$$\begin{aligned}\log(C_{tjk[i]}) = \log(H_{tjk[i]}) + A_i + Y_t + D_j + S_{k[i]} + (YA)_{ti} + (YD)_{tj}\\ + (YS)_{tk[i]} + (AD)_{ij} + (DS)_{jk[i]} + (YAD)_{tij} + (\mathrm{YDS})_{tjk[i]}\\ + \lambda F_{tk},\end{aligned} \tag{3}$$

All terms were treated as random effects except for area (A_i), depth stratum (D_j), their interaction $(AD)_{ij}$ and the depredation effect λ. The addition of $(YDS)_{tjk}$ in Eq. (3) saturates the model, providing an individual random effect for each observation of aggregated catch, C_{tjk}. Such an approach is used to account for overdispersion in Poisson mixed models (Gelman and Hill, 2007). In this context, the variance of YDS reflected natural variation in mean CPUE among year/depth stratum/station combinations, as well as additional overdispersion accrued due to summing catches across skates.

We fit an additional model to evaluate whether sperm whales had differing effects on catch rates by area (ME.2A). This model started with the structure defined in Eq. (3), with the addition of an interaction of area with depredation $(AF)_{itk}$. To examine the sensitivity of model estimates over time, we fit the four models to subsets of historical data ending in years 2004 through 2016 to see if depredation estimates changed markedly with the addition of each year of data.

We conducted simulations to examine the estimates of the different models evaluated in this study when fitted to unbalanced survey data (see Supplementary material for more details). In brief, four combinations for the depredation effect (λ) and lag-1 autocorrelation (ϕ) for year effects (Y_t) were simulated: (1) $\lambda = 0.0$ and $\phi = 0.0$; (2) $\lambda = 0.0$ and $\phi = 0.9$; (3) $\lambda = -0.2$ and $\phi = 0.0$; and (4) $\lambda = -0.2$ and $\phi = 0.9$. For each combination, 1000 datasets (trials) were generated and all four models (QP, NB, ME.1, and ME.2 as described above) were fit to each. These simulations showed that GLMMs provided far superior estimates of depredation effects than the GLMs (see Supplementary material).

All analyses were conducted using R Statistical Computing Software (R Development Core Team, 2017 version 3.3.0). The QP and NB models were fit using the *glm* and *glm.nb* functions, respectively (Venables and Ripley, 2002). The mixed-effects models were fit using the maximum likelihood (Laplace approximation) method of the *glmer* function (as implemented in R package 'lme4'; Bates et al., 2015).

2.5. Stock assessment application

The Alaska sablefish stock is assessed annually by fitting an age-structured model to commercial and survey catch rates (which are treated as indices of abundance) and commercial and survey catch age and length compositions (Hanselman et al., 2016). Catch rate data are treated as lognormal; age and length compositional data are treated as multinomial. Natural mortality and growth schedule (mean and variance of length and weight at age) are estimated externally. The model fit included estimates of present and historical abundance (including annual recruitments), fishing mortalities, fishery and survey selectivities, and catchabilities. Present abundance is projected forward to recommend catch limits (ABCs) for future fishing years, which in turn inform the IFQ limits for the commercial fishery.

The additional sablefish removals associated with sperm whale depredation have not been accounted for in the stock assessment model; therefore, there is likely some negative bias in model outputs such as spawning biomass and recruitment because survey catch rates may be artificially low due to depredation removals. We investigated the effect of the across-areas depredation effect (λ) estimated in model ME.2 (Section 3.2). This provided an expansion factor of (1/exp(λ)) for inflating catches at depredated stations. Using these new survey catches inflated to account for depredation, a revised survey abundance index was used in the sablefish assessment model. We then evaluated the effect of using this depredation-corrected index on estimates of female spawning biomass and other quantities of management interest estimated by the reference assessment model (Hanselman et al., 2016). To examine the implications of uncertainty in the depredation effect (λ), we repeated this procedure using depredation effect sizes corresponding to the lower and upper 95% confidence bounds for λ.

3. Results

3.1. Spatial and temporal patterns

From 1990–2016, data were collected at 942 longline survey year/station combinations across the CGOA, WY, and EY/SE management areas. Since 1998, when observations of depredation began, there were 662 station- and year-specific observations (Table 1). Sperm whales were reported as depredating in 202 cases (31%). The proportion of stations with depredation varied considerably across years and areas (Fig. 2; Table 1), but was generally low for the CGOA area compared to WY and EY/SE. There were significant (P ≤ 0.05) increasing trends over time in depredation among EY/SE stations (Fig. 2).

Table 1
Number of year/station combinations by area from 1998 through 2016 on the annual NMFS Alaska longline survey. "Evidence" corresponds to evidence of depredation (damaged fish) in combination with sperm whale presence.

Area	Total	Evidence	Percent
Central Gulf of Alaska	301	42	14.0%
West Yakutat	152	75	49.3%
East Yakutat/Southeast Outside	209	85	40.7%
Total	662	202	30.5%

Results of a PCA analysis of CPUE data (see Supplementary material) suggested that much of the spatial and temporal correlation in CPUE across stations could be accounted for by simply modeling the mean annual trend in CPUE (e.g., Y(t) in Eq. (2)). However, there was some indication of spatial structure by area in the PCA, and models that included area terms fit significantly better than those without, indicating consistent area-specific differences in CPUE across years. Therefore, subsequent models included the area effect.

3.2. Mixed effects versus fixed effect models

The mixed-effects models provided more plausible, accurate, and robust estimates of depredation effects than the QP and NB fixed-effects models (Table 2). To put these empirical estimates in context, we conducted simulations of the unbalanced survey data with respect to depredation (see Supplementary material) that showed that depredation estimates derived from fixed-effects models were far more variable than for mixed-effects models (e.g., standard deviations of QP and NB estimates across 1000 trials were roughly 10 times greater than for mixed-effects models). Moreover, although QP and NB estimates had high standard errors (SEs), their corresponding 95% confidence intervals failed to include the true simulated effect roughly 15% of the time, instead of the desired 5%. By comparison, depredation estimates for the two mixed-effects models were unbiased and had lower SEs, with confidence interval coverage near the 95% level when there was no autocorrelation in annual CPUE and slightly less (~91%) with high autocorrelation.

The estimates of depredation from QP and NB models using the data in this study are likely too severe to be biologically plausible (a much more extreme change in the time series would be evident). The ME.1 model had the same structure as the QP and NB models, but estimated year, station, and station-year interactions as random effects. However, when fit to simulated data the ME.1 model performed much better and resulted in a very different estimate of the depredation effect. The QP and NB models were very sensitive to the years included in the analysis (Fig. 3). For most years, the QP and NB models showed a severe reduction (~40%), but when the data set ended between 2007 and 2011, the estimates switched to very small effects on catch rates. For 2016, the data set used in this analysis, the most severe catch reductions were estimated for the QP and NB models. Both ME.1 and ME.2 were extremely stable with estimates of about 15–16% regardless of the amount of data included (Fig. 4). Although the ME.1 and ME.2 models provided very similar depredation estimates (Table 2), the ME.2 provided a significantly better fit overall because it accounted for the full structure and overdispersion of the data. For the remainder of the results we discuss the ME.2 model.

3.3. Depredation effect

The across-area estimates for sperm whale depredation indicated a proportional change in CPUE of 0.85 (95% CI: 0.80–0.90), or a 15%

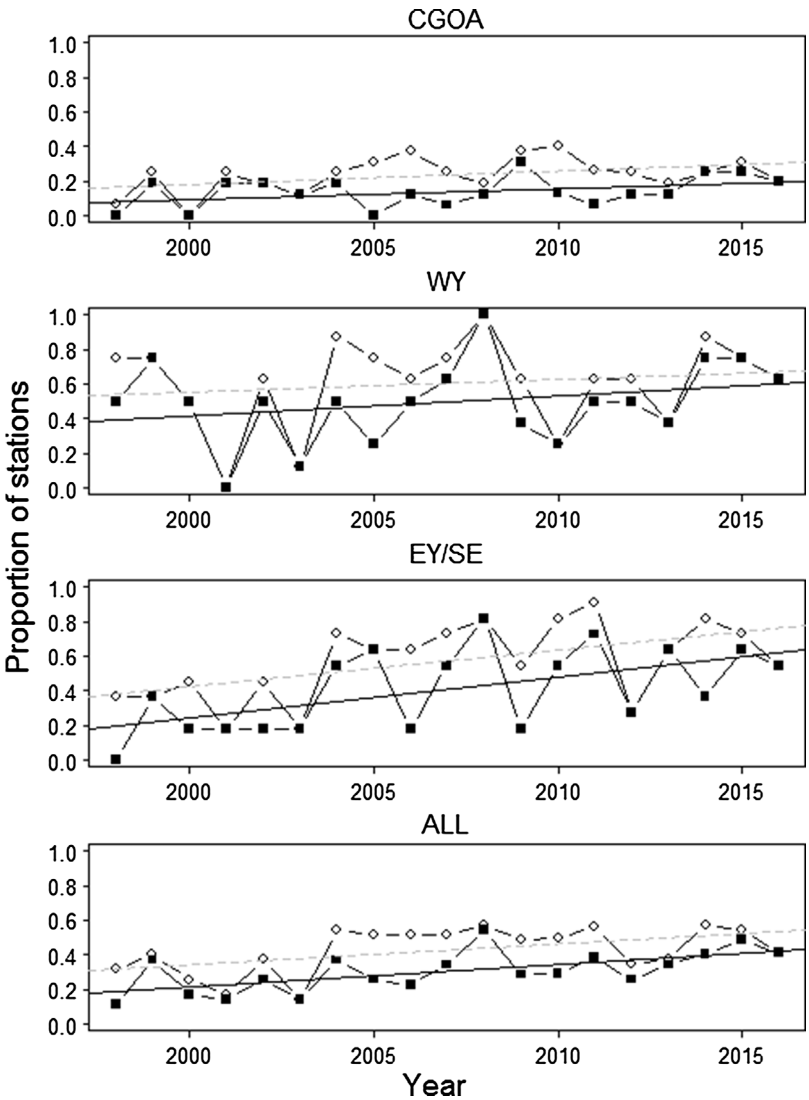

Fig. 2. Proportion of stations with presence (open circles) and evidence of sperm whale depredation (solid squares) by management area and pooled, 1998–2016. Trendlines are linear regressions of presence (grey dashed line) and evidence (black solid line).

Table 2
Estimates of sperm whale depredation for five model configurations. SE = standard error of the estimate. Estimates of proportional change are given by exp(Estimate) with approximate 95% confidence intervals shown (LCI, UCI). For the area model ME.2A, CGOA = Central Gulf of Alaska, WY = West Yakutat, and EY/SE = East Yakutat/Southeast Outside.

				Proportional change		
Model	Estimate (λ)	SE	P value	e^{λ}	LCI	UCI
QP	−1.177	0.392	0.003	0.308	0.143	0.665
NB	−0.692	0.468	0.139	0.500	0.200	1.252
ME.1	−0.183	0.028	< 0.001	0.833	0.788	0.881
ME.2	−0.162	0.031	< 0.001	0.850	0.800	0.904
ME.2A:CGOA	−0.107	0.058	0.062	0.898	0.802	1.005
ME.2A:WY	−0.196	0.054	< 0.001	0.822	0.740	0.915
ME.2A:EY/SE	−0.176	0.050	< 0.001	0.839	0.761	0.924

reduction in sablefish CPUE (ME.2, Table 2). When area effects of depredation were estimated, the ME.2A model had a slightly higher AIC (ΔAIC = + 2.6), suggesting little evidence of strong area-specific effects. Area-specific estimates of proportional change ranged from 0.82 for EY/SE ($P < 0.05$) to 0.90 for the CGOA ($P > 0.05$; Table 2).

3.4. Stock assessment application

Sperm whale depredation was accounted for in the Alaska-wide

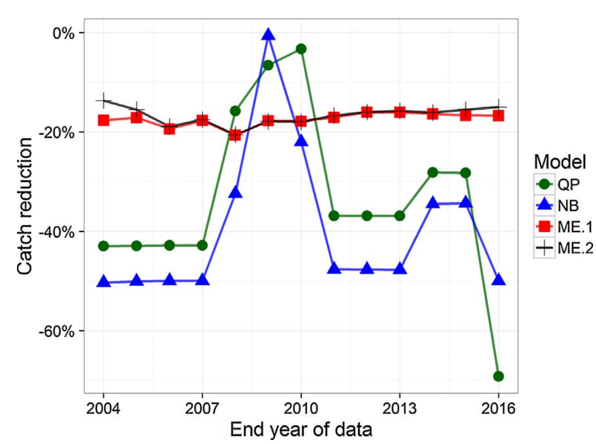

Fig. 3. Estimates of sablefish catch reduction for the four all-area model types, with successive years of data added on starting in 2004. QP = Quasipoisson, NB = Negative Binomial, ME = Mixed Effects.

longline survey index using the estimate of proportion change (0.850; 95% CI: 0.800–0.904) for the ME.2 model (Table 2). This provided an expansion factor of 1.176 (95% CI: 1.106–1.250) for inflating catches at depredated stations. While the effect was substantial at depredated stations, depredation occurred at a small proportion (13% across

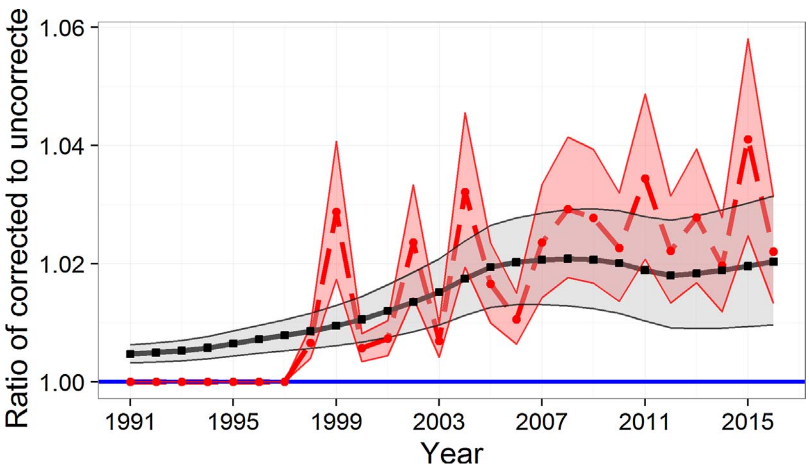

Fig. 4. The effect of accounting for sperm whale depredation on the sablefish relative population number index (dashed red line with filled circles) and estimated female spawning biomass (black solid line with filled squares). Blue line at unity corresponds to the uncorrected index and assessment model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

years) of the total survey stations, which included those in the western GOA, the Bering Sea, and the Aleutian Islands. Consequently, accounting for sperm whale depredation resulted in only modest increases in the all-Alaska longline survey abundance index, ranging between 1% and 4% from 1998 to 2016 (Fig. 4).

Use of the corrected abundance indices in the stock assessment model resulted in an increasing divergence over time in estimates of female spawning biomass (SSB) relative to reference model estimates (Fig. 4). The terminal year estimate of SSB was about 2% greater than the reference model when the adjustment factor for depredation was included (Fig. 4). The current sablefish assessment estimate of female spawning biomass is below the target reference point ($B_{40\%}$), which results in further downward adjustments of recommended ABCs. The depredation corrected assessment model resulted in a relatively larger increase in SSB than the $B_{40\%}$ reference point. Because the sablefish stock is currently below the $B_{40\%}$ reference point, any increase in stock status results in a larger increase in the recommended fishing mortality because of the North Pacific control rule. Consequently, for the depredation corrected model, the increase in available biomass resulted in an increase in recommended ABC (allowable IFQ) that was 3% greater than the reference model (Fig. 5). In addition, the assessment model that corrected for sperm whale depredation provided a better fit to the data, but estimated similar values for average recruitment and survey catchability (Fig. 5). Assessment models based on corrections using the lower and upper intervals for the depredation estimate yielded roughly 1% and 5% increases in recommended ABC, respectively, compared to the reference model (Fig. 5).

4. Discussion

4.1. Spatial and temporal patterns

The proportion of longline survey stations impacted increased significantly from 1998 to 2016 in the CGOA and EY/SE and in all areas when pooled ($P < 0.05$). These trends were largely due to low proportions of impacted stations prior to 2004 (Fig. 2). From 2004–2016, there was little evidence of a linear trend in any data set ($P > 0.1$). Increasing depredation trends in the late 1990s and early 2000s parallel anecdotal trends seen in the commercial longline fishery (Peterson and Carothers, 2013) and shown in Peterson and Hanselman (2017). In 1995, the sablefish and Pacific halibut longline fisheries transitioned to an Individual Fishing Quota (IFQ) management system and extended their fishing season from quick "derby-style" openers to full nine-month seasons. The lengthened sablefish season may have provided more opportunities for sperm whales to learn to depredate on fishing gear and rehearse the behavior (Hill et al., 1999; Peterson and Carothers, 2013; Sigler et al., 2008; Thode et al., 2005). Although no current estimate of sperm whale abundance in the GOA exists (Allen and Angliss, 2013), an increasing number of interactions in most regions suggest that sperm whale numbers may be increasing in some areas. Sperm whales experienced heavy exploitation from commercial whaling in the North Pacific through the late 1970s (Ivashchenko and Clapham, 2014; Mizroch and Rice, 2013), but their numbers may be recovering in the GOA since the cessation of commercial whaling pressure in the 1980s.

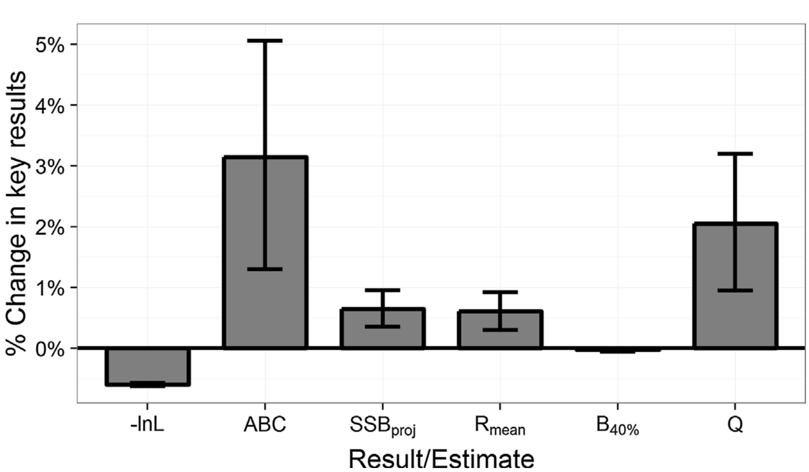

Fig. 5. Change in key results in Alaska sablefish assessment when sperm whale depredation is accounted for in the longline survey; "-lnL" is the negative log likelihood, "ABC" is the Acceptable Biological Catch recommendation, "SSB_{proj}" is the female spawning biomass projected for the following year, "R_{mean}" is the estimated average recruitment, $B_{40\%}$ is the estimated target reference point, and "Q" is the catchability or proportionality constant for the longline survey abundance index.

4.2. Model selection and findings

Results from this study suggest that mixed-effects models were the most promising method for estimating the sperm whale depredation effect. Based on model performance indicators, such as point estimates and standards errors, and results from simulations, the mixed-effects models performed better than the fixed-effects GLMs (Table 2, Table 1 in Supplementary material). This result suggests that the fixed effect structure used in Peterson et al. (2013) that performed reasonably for killer whale data, may be inadequate for sperm whale data in the current context. This is likely due to the fact that the killer whale effect was much higher, and the data were more balanced with most stations having skates with both depredated years and non-depredated years. The mixed-effect model described here would likely provide similar estimates of mean effect size for killer whales as in Peterson et al. (2013), but with better estimates of standard errors. It would be possible to fit a simpler GLM without the interaction terms between year and station. However, we would contend that such a model may provide a reasonable point estimate if the relative variability is low enough at the year-station level, but not accounting for the structure of the data when this variability is substantial could significantly bias the point estimate and standard errors for depredation effects.

Simulations based on the structure of this data set showed that the mixed-effect models provided reliable inferences of depredation effects, even in the presence of significant autocorrelation (see Supplementary material, Fig. 1). This study joins the growing support (Bolker et al., 2009; Garson, 2012; Zuur et al., 2009) that mixed-effects models provide superior inferences than fixed-effects models when data are unbalanced (e.g., areas and stations that have never had whale interactions or stations where sampling did not occur) or contain temporal and/or spatial autocorrelation (e.g., repeatedly measuring the same stations that have spatial covariance). Tixier et al. (2016) also preferred GLMMs to model depredation by killer whales in the Patagonian toothfish fishery because of the unbalanced data.

The use of mixed effect modeling showed that evidence of depredation has a significant and substantial effect on catch rates of sablefish on the Alaska longline sablefish survey in the areas where sperm whales are frequently encountered. Evidence of depredation was associated with an estimated reduction in sablefish CPUE of 15% (Model ME.2, Table 2). The sperm whale depredation effect varied by area when area effects were estimated in the across-area model, but the model had a slightly higher AIC ($\Delta = 2.6$) and not all of the area depredation effects were significant (Model 2A, Table 2). While the frequency of depredation events may differ between areas, it is reasonable that the effect of depredation on sablefish CPUE (e.g., at a given station) would be consistent among areas because the same whales are often observed depredating across areas (Straley et al., 2014). Nevertheless, area-specific differences in depredation effects from west to east could occur due to differences in sperm whale diet and ecology. The weakest depredation estimate was for the western CGOA region (Table 2), and estimates of the depredation effect in the fishery show a similar, but nonsignificant, increase from west to east (Peterson and Hanselman, 2017). Diet information from historic sperm whale catch data indicates sperm whales caught in the Bering Sea and Aleutian Islands relied more on cephalopods, while fish became a more significant component of sperm whale diet towards the eastern Aleutians and Gulf of Alaska (Okutani and Nemoto, 1964; Sigler et al., 2008). However, this cline could also be an artifact of depredation being a more recent phenomenon in the CGOA.

This research builds upon previous studies that quantified the effect of sperm whales on fishery and survey catch rates in Alaska. To estimate sperm whale depredation on the AFSC longline survey, Sigler et al. (2008) fit a repeated measures model to the longline survey data and found neither sperm whale presence nor catch removals increased significantly from 1998 to 2004. Catch rates were estimated to be about 2% less at station locations where depredation occurred, but the effect was not significant ($P > 0.05$). A weakness of Sigler et al. (2008) study was that it fit a Gaussian distribution to sablefish weight, which is a derived quantity based on measured lengths expanded across the station, while our current analysis assumes a more appropriate Poisson distribution for the true count data (numbers of sablefish). Despite being much larger than the effect estimated by Sigler et al. (2008), using numbers may still be a conservative estimate if sperm whales were targeting larger sablefish. However, there is no evidence that this is occurring on the survey. Another study evaluating data collected in southeast Alaska, found a significant depredation effect when longline fishery catches between sets with and without sperm whales present were compared (3% reduction, 95% CI of (0.4–5.5%, *t*-test, $p = 0.02$; Straley et al., 2005). Their study used sperm whale "presence" as an indicator during the years 2003 and 2004 on commercial fishing vessels in Southeast Alaska. Results from our study suggest much higher percentage CPUE reductions when sperm whales are depredating (15%) than these previous studies in the Alaska region. The higher CPUE reductions estimated in our analyses were likely a product of the longer time series associated with the current study (1990–2016) in tandem with the refined modeling approach that incorporated random effects and including the full interaction structure. Sperm whale depredation was relatively more frequent and consistent from 2004 to 2016, and additional data coupled with mixed models may have yielded a stronger depredation signal in the survey catch data. We also showed that not modeling the full structure of the data with mixed effects can result in widely variable effect size estimates, depending on which years were modeled (Fig. 3).

This study focused on improving upon previous GLM approaches used in Alaska and it has previously been suggested that GLMMs were a superior method for estimating the effects of depredation (Tixier et al., 2016). Alternative analyses that could have been investigated for this study include General Estimating Equations (GEEs), and Bayesian implementations of the methods explored in this study. General Estimating Equations (Hardin and Hilbe, 2003) are a useful approach when interested in population level parameters (as we eventually employed in the stock assessment application here) and are one way of imposing a correlation structure on longitudinal data (i.e., repeated measurements at fixed stations) such as these. Bayesian implementation of GLMM has been suggested (Fong et al., 2010). GEEs and the Bayesian approach would be useful in future explorations, including basing prior distributions of the depredation effect estimates that have been based on other data and comparing the computationally more convenient GEEs to more complicated Bayesian GLMMs.

4.3. Stock assessment implications

The mixed effects model (ME.2) yielded the most reliable estimates of depredation in this study and were used to quantify CPUE reductions (λ) due to sperm whale depredation and inflate catch estimates at stations of the AFSC longline survey. This new abundance index was input into the reference sablefish assessment model (Model 10.3b, Hanselman et al., 2016). Inclusion of the depredation corrected abundance indices resulted in a recommended ABC that was 3% greater than the reference model (Fig. 5). In a valuable ($/lb) stock such as Alaska sablefish, a 3% increase in ABC translates to nearly US $3 million (Fissel, 2014) in increased revenue.

There are two caveats to consider before utilizing these corrected indices in the sablefish stock assessment. First, the extent of sperm whale depredation prior to 1998 in the survey is unknown. Considering the apparent increase in sperm whale depredation in recent years, it is likely that depredation historically had a minor impact, but it is an added uncertainty. Second, accounting for the additional mortality that whales cause in the fishery is a critical step before using corrected abundance indices to increase the recommended quota from the stock assessment, because current estimates of catch are lower than the true number of sablefish killed, but not successfully harvested. Fortunately, estimates of the amount of sablefish consumed by whales in the fishery have been recently published (Peterson and Hanselman, 2017).

4.4. Management implications

This study is especially relevant in light of recent management actions implemented by the North Pacific Fishery Management Council (NPFMC) to legalize pot (trap) fishing gear for sablefish in the GOA. Due to an increasing number of reports and testimony by fishermen concerning whale depredation in the GOA, the NPFMC passed an amendment in June 2015 (http://www.npfmc.org/halibutsablefish-ifq-program/http://www.npfmc.org/halibutsablefish-ifq-program/) to allow pot fishing gear in the region. This analysis helped inform this action, and the refined estimates of the sperm whale effect on CPUE suggest that sperm whale depredation has an impact on commercial harvests and survey abundance indices. Sablefish caught in pot gear are not available to whale depredation; therefore, with the approval of the measure, fishermen will be able to switch to pot fishing gear to reduce lost catch to depredation beginning in 2017, but it is likely that longline fishing will also continue.

In contrast to the commercial fishery (e.g., Rabearisoa et al., 2015), NMFS has fewer options for preventing whale depredation on the survey. The survey must fish specific locations at certain times of the season irrespective of the presence of depredating whales. Deterrence measures are not allowed under current U.S. laws, (e.g., the Marine Mammal Protection Act). Additionally, the long-term (1979-current) stock assessment time series of the standardized longline survey is critical for robust results. By switching survey gears (from longline to pots/traps), whale depredation could be prevented, but the benefits of a long abundance time series would be reduced and large-scale calibration studies would be required prior to using a new time series of survey data with different catchability and selectivity characteristics. This study makes some progress to alleviate stakeholder concerns that the current survey is becoming less accurate.

4.5. Conclusions

This study represents a significant step towards better accounting for the effect of sperm whale depredation on the sablefish index of abundance from the AFSC longline survey. Inclusion of a larger data set in this study, along with the improvements of using mixed-effect models resulted in a much larger effect of sperm whale depredation than has previously been reported in Alaska. The results from this analysis could be re-estimated every few years to ensure they are not changing over time from the estimate used to correct survey indices, either due to learning behaviors or spatial changes in whale distributions relative to sablefish abundance. However, the estimates appear quite robust to the addition of new data thus far. With the results of this study and the recent estimation of total mortality caused by whales in the fishery (Peterson and Hanselman, 2017), the two components necessary to appropriately account for whale mortality in the stock assessment are available. Incorporation of these results should lead to a more accurate assessment and improved future management of the important Alaska sablefish resource.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.fishres.2017.12.017.

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